



Flight Traffic Analytics using MongoDB

Overview

This project was built as part of my MongoDB learning journey.

The goal of this project was to gain practical experience with MongoDB queries, aggregations, indexing, and performance optimization using a real-world flight traffic dataset.

Instead of focusing only on theory, I worked directly with flight records and solved analytical problems using MongoDB Query Language and the Aggregation Framework.

Dataset

The dataset contains flight traffic information including:

- Airport Name
 - Airport Code
 - Country
 - Flight Date
 - Year
 - Month
 - Arrivals
 - Departures
 - Total Flights
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What I Practiced

Basic Querying

- Exact Match Filters
- Multiple Condition Filters
- Comparison Operators
- Inclusion Queries using `$in`

Aggregation Framework

- `$group`
- `$project`
- `$match`
- `$sort`
- `$limit`

- \$avg
- \$sum

Performance Optimization

- Explain Plans
 - Compound Indexes
 - Query Performance Analysis
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Project Tasks

The project includes:

- Country-based flight analysis
 - Date-based filtering
 - Airport-specific searches
 - Traffic volume calculations
 - Country-wise aggregations
 - Monthly traffic analysis
 - Departure and arrival diagnostics
 - Query optimization using indexes
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Repository Structure

```
mongodb-flight-traffic-analysis/
```

```
├─ README.md
├─ flights_dataset.json
└─ mongodb_queries.md
```

My Learning Outcomes

Through this project, I learned:

- How MongoDB stores data using documents
 - How to query real-world datasets
 - How aggregation pipelines work
 - How to reshape and analyze data
 - How indexing improves performance
 - How to read and understand explain plans
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Notes

The queries included in this repository represent my own learning and practice work while studying MongoDB concepts and solving dataset-based exercises.

Technologies Used

- MongoDB
 - MongoDB Compass
 - JSON Dataset
 - Aggregation Framework
 - Indexing
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Author

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